Further applications of the normal d[redacted]ution

1

a 44% b 89% c $50\,848$

2

a 23% b 13% c 2524

3

a 0.769 b 70.060% c 31.757%

4

a 26.154 cm b 27.280 cm c 24.724 cm d 0.248

e 0.690

5

a 3.493 kg b 3.721 kg c 3.363 kg d 0.309

e 0.614

6

a 190.047 cm b 195.869 cm c 4.8% d 0.566

e 0.371

7

a $5.108, 5.012$

b 0.3

c There is a good chance that the machine is faulty.

8

a 301.15 K b 3.3 K

9

a 3500 g b 550 g

10

a 550 mm b 1600 mm

11

a $b = 4$ b 68.48

12

a -328

b 16

13

a 2.985

b 67.848 g

14

a 34.457 cm

b $(X+11)$ cm

c 34.457 cm

15

a $0.706=\dfrac{70-\mu}{\sigma}$

b $-0.151=\dfrac{56-\mu}{\sigma}$

c $\mu=58.465, \sigma=16.331$

16

a 58.7%

b $(0.7\mu+2)\%$

c $(0.7\sigma)\%$

17

a 122.736

b $\dfrac{20}{37}X$

18

a 40%

b 3

c 69%

19

a 40.1

b 0.69

c 0.035

d 0.346

e 1.74 mins

20

a 83%

b 0.64

c 44%

21

a 0.43

b 172.83 g

c 0.85

d 177.26 g